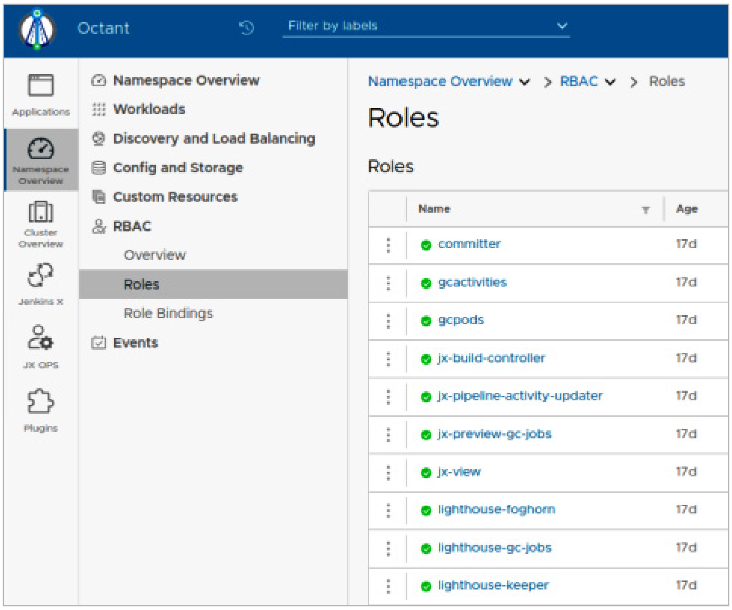

Figure 11: Roles in JX

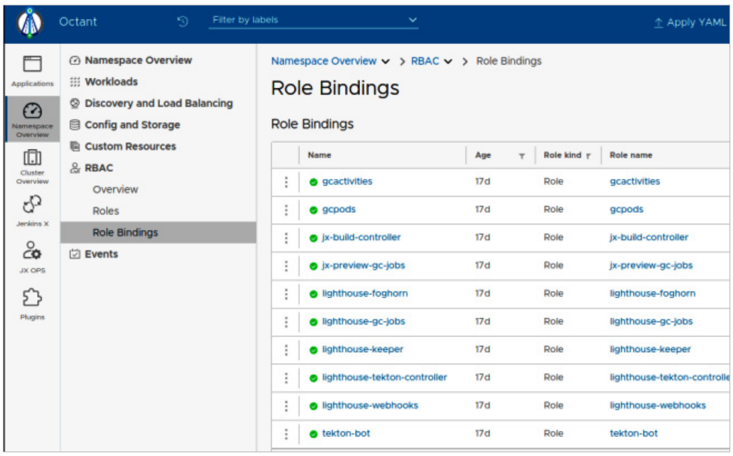

Figure 12: RoleBindings in JX

7. Check all the components created in k8s (pods, deployments, job, etc) by running the following command:

```
$kubectl get all
```

Getting the dashboard

In order to use the Octant UI, you must be in a local terminal (as opposed to a virtual machine) since it will launch a local process and open a web browser to access a local port. Type the command:

```
$ jx ui
```

Overview: Figures 6 and 7 show the initial overview screen in Octant, where users can get an overview of Jenkins X.

Discovery and load balancing: The screen shown in Figure 8 displays Octant's built-in settings and pipeline repositories, along with load balancing and discovery features.

Ingress is an API object that provides HTTP and HTTPS routing to services based on rules. It allows external access to services within a Kubernetes cluster. Ingress controllers are responsible for implementing the rules specified in the Ingress resource and managing external access to services.

Services: Service is an abstraction that defines a logical set of pods and a policy by which to access them. It enables external traffic to reach a set of pods in a consistent manner.

Deployments: A deployment is a resource that defines a desired state for a set of replica pods. It allows you to describe the characteristics of your application, such as the number of replicas, the Docker image to use, and how to update it. Deployments provide a way to manage the rollout and rollback of application updates.

ConfigMaps: Configuration artifacts and containerised apps are separated by ConfigMaps. They let you mount non-sensitive configuration data into containers as files in a disk or as environment variables by storing it in key-value pairs.

Secrets: Similar to ConfigMaps, secrets are used to store sensitive information, such as passwords, API keys, and tokens. Secrets encode and store this information in a way that ensures confidentiality. They can be mounted into containers as either environment variables or as files in a volume.

Roles and RoleBindings

- ***Role:*** A Role is a Kubernetes resource that defines a set of permissions (verbs) within a specific name space. It allows you to specify what actions a user, group, or service account is allowed to perform on specific resources in that name space.
- ***RoleBinding:*** A RoleBinding binds a Role to a user, group, or service account, specifying the set of permissions granted. It establishes the link between a Role and the entities that should have those permissions within a name space.

By following these installation steps, Jenkins X can be installed successfully and made operational. You can now develop your project and actively engage with it. END

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